

Name: Solar Panel Team**Date:** 9/26/2025**Class:** Design for Engineers e4usa

MyDesign Portfolio

Element A: Presentation and Justification of the Problem

Problem Statement and Details (A1, A2, A5)

In the U.S., non-renewable sources of energy like coal, natural gas, petroleum, or oil, account for 83% of the United States energy resources (U.S. Energy Information Administration, 2023, "U.S. energy facts explained"), but using these sources contributes to the United State's 6.3 billion metric tonnage of carbon emissions annually, one of the largest in the world (United States Environmental Protection Agency, 2025, "Sources of Greenhouse Gas Emissions"). To address the climate and scarcity issues caused by the use of these nonrenewable energy sources, modern society is shifting towards energy resources that are cleaner and more renewable than fossil fuels, like hydroelectric plants, nuclear plants, wind energy, and solar energy. We're focusing on solar energy and the problem of solar panel efficiency. Even the most modern and well-constructed solar panels (that are also the most expensive) only achieve about 40% efficiency, meaning just 40% of the sun's light that hits the panel becomes usable energy. The rest is wasted in reflected light, heat irradiation, and losses during photoelectric conversion processes. The average solar panel efficiency is a lot lower, hovering around 20% (University of Michigan Center For Sustainable Systems, 2022, "Solar PV Energy Factsheet"), as cheaper and less efficient panels become popular in markets. Solving the problem of solar panel efficiency is an integral step in making sure that these renewable sources of energy have the ability to become the main supply of energy to society in the future, addressing problems of grid diversity, climate change, and energy insecurity.

Stakeholders involved in our solar panel solution include energy companies, local business owners, homeowners, and even the government. Each stakeholder has the ability to use solar panel innovation to make energy systems more efficient. It is important to recognize the role climate change plays in solar panels, as sustainable solar energy combats rising greenhouse gas emissions from nonrenewable energy



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sources. This solar panel situation is unique because it can vary in scope. While renewable energy and climate change are global issues, it is important that we focus on changes at the local level because implementing these is more feasible and realistic. Solar panel inefficiency occurs within the solar panel as it absorbs the sunlight. This problem occurs whenever the solar panels are active. This problem occurs because of the physical limitations of solar panels, energy transfer loss, and environmental factors. This problem can occur in any solar panel system, and the different factors that contribute to it. This makes finding our exact parameters for finding the right solution important to keep the scope of the project in reach.

In addition to surface debris, solar panels also suffer from degradation due to heat, humidity, and voltage. Hot spots and microcracks can further decrease efficiency and potentially pose fire risks over time. These issues collectively shorten the lifespan of panels and increase the long-term cost of solar energy, which can be mitigated by a prototype that cleans and prevents debris buildup. Since the problem is so widespread, many areas in the country are trying to find ways to innovate. However, this is having some unintended consequences. For example, the amount of photovoltaic waste generated by solar panels could contaminate soil and water with hazardous substances like lead and cadmium. Another issue is that manufacturing these panels also causes toxic substances to be released into the air, which leaves workers in the factories at risk of exposure to these chemicals.

Stakeholder Concerns and Connections (A3, A4)

1. Energy companies are a primary stakeholder because they are responsible for supplying residents and businesses with solar panels. As a result, they are always looking for ways to deliver reliable energy to their customers. We can contact the energy companies through their business representatives.
2. Business owners are a primary stakeholder because many of them are shifting away from fossil fuels and starting to use cleaner energy like solar panels. This is because they want to reduce the energy costs of their business while making the environment healthier. For example, local businesses with solar panels can be contacted with our information.



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3. Residents are a primary stakeholder because many of them are shifting away from fossil fuels and starting to use cleaner energy like solar panels. This is because they want to reduce the energy costs of their houses while making the environment healthier. After our tests, our prototype will be displayed to consumers who are interested in solar energy.
4. The government is a secondary stakeholder because it makes some money from the companies that sell solar panels. They also use subsidies and tax credits in order to make these companies produce more or fewer solar panels. We can contact the government through local grassroots petitioning.

Problem Importance (A6)

The efficiency of a solar panel decides how much sunlight it converts into electricity. Most panels on the market today convert only 15 % to 22 % of the light that strikes them (Department of Energy, 2025). Because of that low rate, more modules, more land, and more capital are required to cover a given power demand. Simple maintenance also changes the output - washing the surface once can lift the energy yield by up to 25 % (Saffarini, 2025). Unless those limits are removed, solar power will stay behind fossil fuels in delivered energy, even though it produces no direct emissions. Higher panel efficiency is urgent for climate targets and for the shape of tomorrow's grids. NASA has been testing larger, tougher arrays for deep-space craft, a sign that the technology matters beyond Earth (NASA, 2024). These efforts show that steady gains in panel performance are required if cleaner and more sustainable electricity is to become the norm.

Reference List: Should have in MLA format the reference you cited

Dada, M., and P. Popoola. "Recent Advances in Solar Photovoltaic Materials and Systems for Energy Storage Applications: A Review." *Beni-Suef University Journal of Basic and Applied Sciences*, vol. 12, no. 66, 2023, <https://doi.org/10.1186/s43088-023-00405-5>.

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Department of Energy. "Solar Performance and Efficiency." *Energy.gov*, 2025, www.energy.gov/eere/solar/solar-performance-and-efficiency.

Elliott, David. "The Breakthroughs Making Solar Panels More Efficient." *World Economic Forum*, 22 Oct. 2024, www.weforum.org/stories/2024/10/solar-panel-innovations/.

NASA. "How NASA Uses and Improves Solar Power." *NASA*, 22 Oct. 2024, science.nasa.gov/sun/how-nasa-uses-and-improves-solar-power/.

Roberson, Mary-Russell. "Building a Better Solar Cell." *Duke Science and Technology*, dst.duke.edu/our-work/materials-science/building-a-better-solar-cell/. Accessed 24 Sept. 2025.

Saffarini, Ryan, CTO. "Maximize Your Solar Panel Production: Tips and Insights." *Sun Valley Solar Solutions (SVSS)*, 24 Mar. 2025, www.sunvalleysolar.com/blog/maximize-solar-panel-production.